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The Effect of Peer Separation in Young Chimpanzees *(Pan troglodytes)*

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Ten chimpanzees (*Pan troglogytes*), aged 18–24 months, housed without mothers as two dyads and two triads, were subjected to social separation. Two issues were addressed: the effects of peer separation in chimpanzees; and differential responses by subjects living in dyads compared with those living in triads. Chimpanzees that were alone during separation reacted with high levels of "protest" alternating with "despair" throughout the separation period. The continued presence of one cagemate, during separation from a third, was a strong mitigating factor. Even when the primary attachment was formed with the absent cagemate, the remaining chimpanzees clung to each other and the levels of protest and despair, when present, were low. Upon reunion, neither "detachment" nor heightened levels of clinging were conspicuous, but there was increased social interaction. The data on separation of chimpanzees are intermediate between those of humans and monkeys separated from mothers or peers. The increased social interactions during reunion, including looking, are comparable to the visual vigilance reported for humans.

Key words: peer separation, attachment, chimpanzees

INTRODUCTION

The potentially deleterious effects of social separation were brought to scientific attention by Spitz [1946], who reported incidences of catatonia, developmental regression, and retardation in institutionalized children separated from their mothers. He labeled this phenomenon "anaclitic depression." Bowlby [1960, 1961, 1969, 1973, 1979] and Robertson [1953] extended this work by providing a general theoretical perspective that also applied to brief separations in noninstitutionalized children. The investigation of analogues of anaclitic depression using nonhuman primates began with Harlow's pioneering studies, which experimentally induced psychopathology in rhesus monkeys [Seay et al., 1962; Harlow & Harlow, 1965]. Most quantitative data in this area is the result of experimental work with monkeys [eg, pig-tailed macaques: Kaufman & Rosenblum, 1967; Reite, 1977; bonnet macaques; Rosenblum, 1971; Kaufman & Stynes, 1978; rhesus macaques: Suomi et al, 1970; Mitchell et al, 1979; patas: Preston et al, 1970]. The great apes, phylogenetically closer to humans than are monkeys, have not been used in these investigations until recently [gorilla: Nadler & Green, 1975; Hoff,

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et al, 1979; orangutans: Nadler & Codner, in press]. Additional data comparing species closely related to humans are clearly needed.

Social separation, defined as "the restriction of access to an individual (or individuals) with whom a subject had previously established a social relationship . . ." [Mineka & Suomi, 1978, p 1376], has been used as a technique to identify the presence and quality of attachment, primarily between a mother and infant. The reaction to social separation in monkey infants consists of a biphasic pattern of behavior. "Protest," characterized by excessive vocalizations and high levels of disoriented locomotion, occurs upon separation and can last up to 36 hours in pig-tailed macaques [Rosenblum, 1971] or 48 hours in rhesus macaques [Mineka & Suomi, 1978]. This phase can be followed by a more prolonged phase of "despair," which is characterized by social withdrawal, a decrease in activity, and an increase in self-directed behaviors. Prolonged ventroventral contact and maintenance of proximity, behaviors indicative of heightened attachment, are observed upon reunion with the mother [Rosenblum, 1971; Mineka & Suomi, 1978]. Children separated from their mothers usually demonstrate behaviors classifiable as protest and despair. Protest occurs on the first day and alternates with despair throughout the separation period [Robertson, 1953; Bowlby, 1960; Heinicke & Westheimer, 1965]. Upon reunion, a curious phenomenon, "detachment," is sometimes observed, in which the child initially shows avoidance or hostility toward the mother.

Most studies of attachment are concerned with mothers and infants. Clearly, other close social relationships, with comparable attachment bonds, are possible. Peers can be primary objects of attachment in monkeys [Suomi et al, 1970; Suomi, 1973; Erwin et al, 1974] and humans [Freud & Dann, 1951], demonstrating both protest and despair upon separation, and in monkeys, increased clinging upon reunion. In fact, it has been proposed that a peer-peer paradigm may be more useful than the mother-infant paradigm in investigations of attachment [Suomi et al, 1970].

The present study was designed to address two issues: the effects of peer separation in young chimpanzees; and the differential separation reaction of chimpanzees living in dyads and triads.

METHODS

Subjects

Ten infant chimpanzees *(Pan troglodytes)* in the nursery of the Yerkes Regional Primate Research Center were observed. The infants were housed in four cages in groups of either two or three as indicated in Table I.

Setting

The home cage for each group consisted of two interconnected individual compartments, each of which was approximately 70 cm on each side. Each cage contained two or three small plastic objects (one per individual) in circular, square, and bowling-pin shapes. A length of small-linked chain, approximately 15 cm long, was hung from the ceiling, one in each compartment.

Procedure

The three consecutive phases—baseline, separation, and reunion—lasted two weeks each. During separation a solid metal partition was inserted between compartments, dividing the cage in half. In the cages containing three infants, one infant was separated from the other two; in the cages containing two infants, each was separated from the other. The barrier prevented physical and visual contact between cagemates, but the separated chimpanzees could still hear and smell each other, and could, on alternate days, see other chimpanzees housed in the same room. Thus the infants were not in total social isolation. During reunion the solid barrier was removed, and the individuals

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TABLE I. Subject Information

Subjects	Length of time with mother	Length of time with peer	Age at testing	Sex
Cage 1				
S1 Tika	<1 day	5 mo	1 yr, 11 mo	Female
S2 Arctica	18 days		1 yr, 6 mo	Female
Cage 2				
S3 Joye	<1 day		1 yr, 8 mo	Female
S5 Sabrina	<1 day	6 mo	2 yr	Female
S4 Puddin	3 days		1 yr, 10 mo	Male
Cage 3				
S6 Sara	<2 mo	1 yr	1 yr, 11 mo	Female
S7 Nathan	<1 day		1 yr, 8 mo	Male
Cage 4				
S8 Hallie	<1 mo		1 yr, 11 mo	Female
S9 Betsie	<1 mo	1 yr, 1 mo	1 yr, 11 mo	Female
S10 Artifact	<1 mo		1 yr, 11 mo	Male

were allowed to interact again. For one dyad, two additional phases were included to test the effect of decreased space independent of separation. After reunion was complete, both chimpanzees were placed together on one side of the cage with the barrier installed for two weeks. A second period of two weeks followed, in which the subjects again had access to the entire cage.

Infants were observed in random order 30 minutes per day, five days per week, between 0930 hr and 1130 hr. Behavior was point sampled every 15 seconds. The behavioral categories are defined and presented in Table II. Those points at which the subject was stationary and not engaged in exploration or social activity were recoded as "inactive," to better reflect the passivity which characterized monkeys during despair [Suomi et al, 1970].

Interobserver Agreement

Interobserver agreement was assessed with a second trained observer for one videotaped and two live sessions. Agreement was assesed in two ways: percentage agreement and Cohen's Kappa, a statistic which corrects for chance agreement. A problem with percentage agreement is that is does not correct for agreement due simply to change, and therefore, the obtained score may be inflated [Hollenbeck, 1978]. A problem with Cohen's Kappa is that when all categories but one occur infrequently the Kappa may be very low although percentage agreement is quite high. The Kappa and percentage agreement scores that were obtained for each group of categories are listed in Table II.

RESULTS

Data Reduction and Analysis

To simplify data presentation, the categories of locomotion, stereotypy, distress vocalizations (hoo, whimper, and scream), distress facial expressions (cry face and pout face), and temper tantrum were collapsed into one category labeled protest. The cat-

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TABLE II. Definitions of Behavioral Categories

Category	Definition
Other-directed	Kappa scores: .9500, .8902, .8875 (% agreement: .9300, .9300, .9400)
Play	Physical contact consisting of wrestling, poking, mild biting, and tickling, which may include a play face and panting
Look	Facial orientation to another chimpanzee without physical contact
Groom	Action directed to another individual which includes picking, manipulation of the body surface, and frequently includes close visual regard with coordinated lip-finger movements
Cling	In physical contact with another chimpanzee with at least one limb encircling and/or grasping the others' body
Touch	Physical contact that is not cling, tandem walk, groom, or play, which includes passive contact and the contact of another with hand or arm
Object-directed	
Object Manipulation	Kappa scores: .9809, 1.0000, .8251 (% agreement: .9900, .9800, 1.0000) Activity directed toward movable objects with either the mouth, fingers, or toes
Explore environment	Kappa scores: 1.0000, .5935, .6630 (% agreement: 1.0000, .9900, .9800) Activity directed toward nonmovable aspects of the environment
Self-directed	Kappa scores: .9290, .7402, .9139 (% agreement: .9900, .9600, .9600)
Self-directed	Behavior that is directed at the chimpanzee's own body consisting of autogroom, self-manipulation, and/or hit
Self-cling	Encircling the body with at least one limb; may include the grasping of hair
Posture	Kappa scores: .9567, .7509, .8594 (% agreement: .9800, .9800, .8700)
Locomote	Includes walk alone, swing, and climb. Swing is defined as supporting the body from the ceiling of the cage while moving in an arc. Climb is defined as being in the act of moving up or down the sides of the cage
Stationary	Hanging, sitting, standing, or lying down
Stereotypy	Repetitive motions sometimes idiosyncratic but usually rocking either forward/backward or side/side. Eye-pokes and digit sucks are recorded separately
Tandem walk	Walking in physical contact with another chimpanzee who is also locomoting, usually with an arm draped over the back
Display	Juvenile components include bipedal swagger, foot stomping, banging, dragging, or throwing objects
Distress	[From Van Hooff, 1967; and Van Lawick-Goodall, 1968].
Hoo	Soft, low-pitched single sound
Whimper	Series of "hoo" sounds
Cry face	Mouth wide open with teeth showing and slightly apart
Pout face	Lips protrude, small opening in the center; mouth corners pressed together
Scream	Loud shrieking
Temper tantrum	Frantic body movements, throwing self in the air, against the ground, etc, with full vertical and horizontal retraction of the mouth
Urinate/ defecate	Common usage
Eyes closed	Lids closed at least three fourths of the way over the eyes
Eat	Common usage

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egories of inactive, self-directed behavior, and self-cling were averaged to form the category of despair. Both types of responses have been similarly described and classified by previous researchers [see Suomi et al, 1970].

Each behavioral category, for each subject, was analyzed for changes that occurred during separation. Parametric tests, such as analysis of variance, were considered inappropriate due to the small sample size (ie, less than ten entries per cell). Moreover, the question was whether or not a subject's behavior changed during the days of separation relative to its baseline behavior. Thus, the mean and standard deviation were computed for each subject's baseline, and a confidence interval (CI) was constructed. An observed percentage point was concluded to be significantly different from baseline when that data point fell outside the 95% CI constructed about the mean. Additionally, a CI was computed for the baseline mean to compare with the mean of points during reunion. The reunion mean was concluded to be significantly different from the baseline mean when the reunion mean fell outside the 95% CI for means of samples of the same size.

Reaction to Separation

First day—alone. All six individuals who were alone during separation (individuals from the dyads and isolated members of the triads) exhibited significantly greater amounts of protest in the first session of the first day than during baseline (Table III). The increases in distress vocalizations and facial expressions concurrent with either rapid locomotion or high-velocity rocking ranged from three to 14 times higher than baseline. Typically, the frequency of protest was highest for the first session and dropped thereafter, but remained significantly increased even for the third session of the first day. Tika and Joye (Fig. 1a,b) typify this pattern, while Artifact and Arctica (Fig. 2ab) illustrate the range of individual variability.

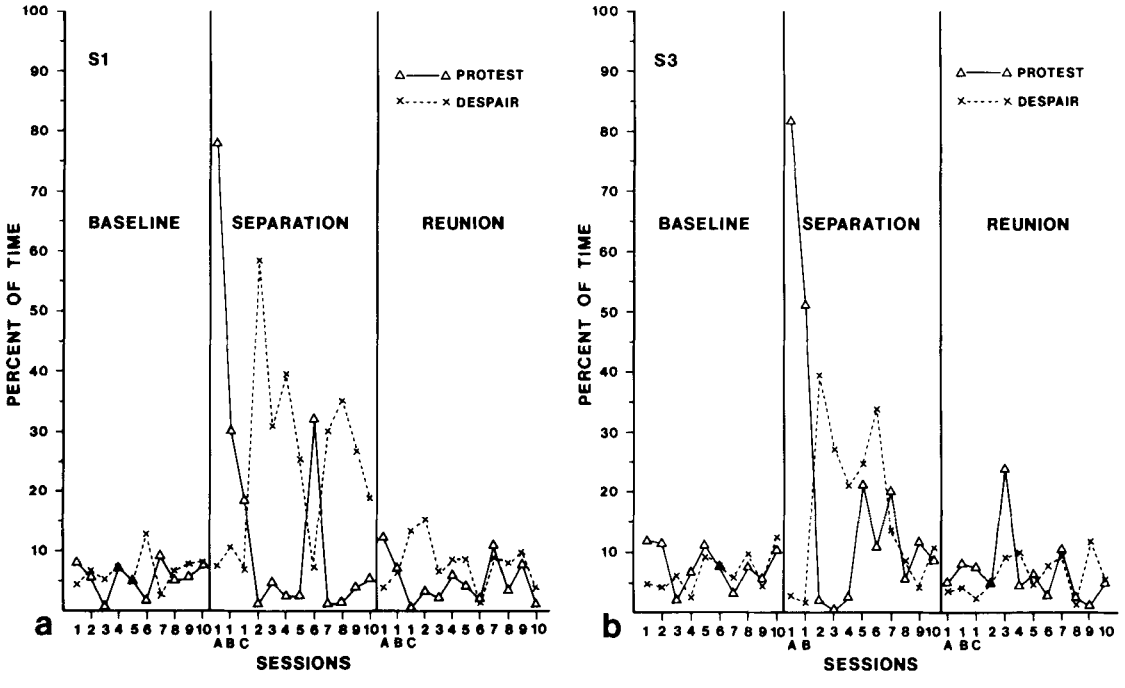


Fig. 1. The protest and despair reactions for Tika (a) and Joye (b).

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Half of the subjects (Figs. 2b, 3a,b) also exhibited significant amounts of despair during some or all of the sessions of the first day of separation. The increases in despair ranged from 1.4 to 4.5 times greater than the amount in baseline. These infants spent considerable amounts of time moving about and vocalizing (ie, protesting), which alternated with periods of sitting quietly, either engaged in self-directed behavior or simply listless (ie, despair).

First day—together. Three of the four chimpanzees from triads who remained with one cagemate during separation exhibited significant levels of protest (over baseline) on the first session of the first day of separation (Table III). The levels of protest, however, were low in comparision with the levels exhibited by the subjects which remained alone during separation; there was at most a threefold increase over the baseline level (Figs. 4a, 5a,b). Affective distress (vocalizations and facial expressions) accompanied protest in only one of these subject, Sabrina, at significant but very low levels (less than 5%). This same individual was the only one who continued to demonstrate increased amounts of protest in the second session. In addition, three of the four "together" subjects spent a significantly increased amount of time clinging to their remaining cagemates.

Despair was evident in only one of these "together" subjects [Hallie] on the second session of the first day, at a level slightly more than twice her average baseline level.

Days 2–10—alone. All six chimpanzees exhibited significantly increased amounts of despair on four to eight of the last nine days of separation. Four of the six subjects continued to show increased levels of despair on the last day of separation.

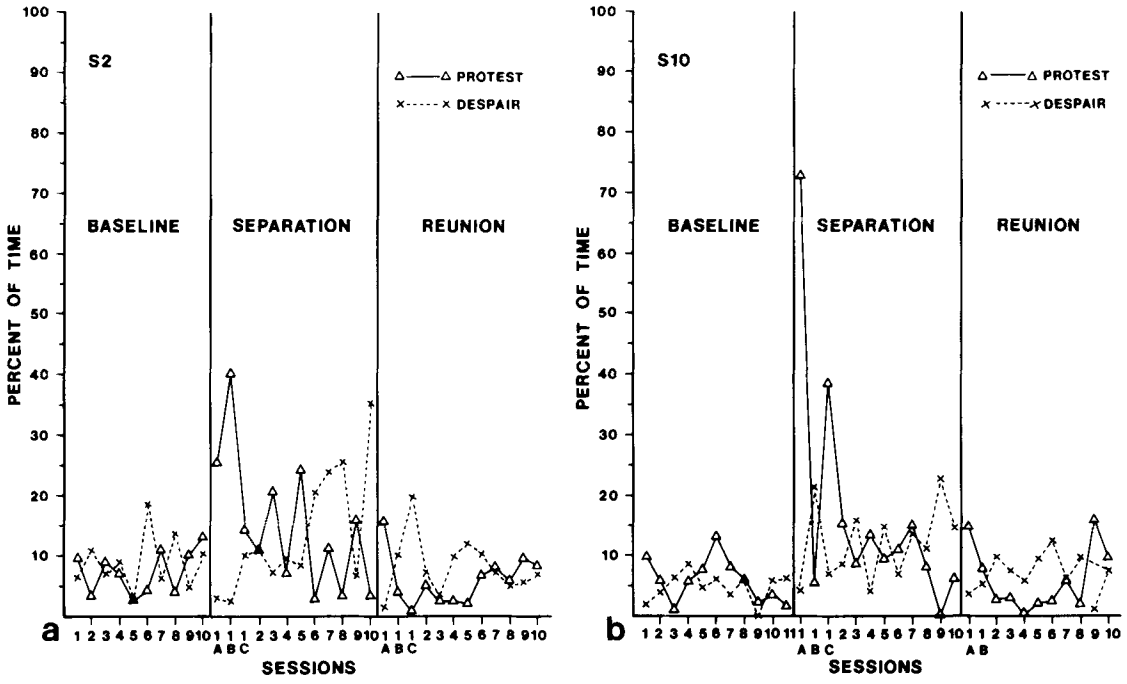


Fig. 2. The protest and despair reactions for Arctica (a) and Artifact (b).

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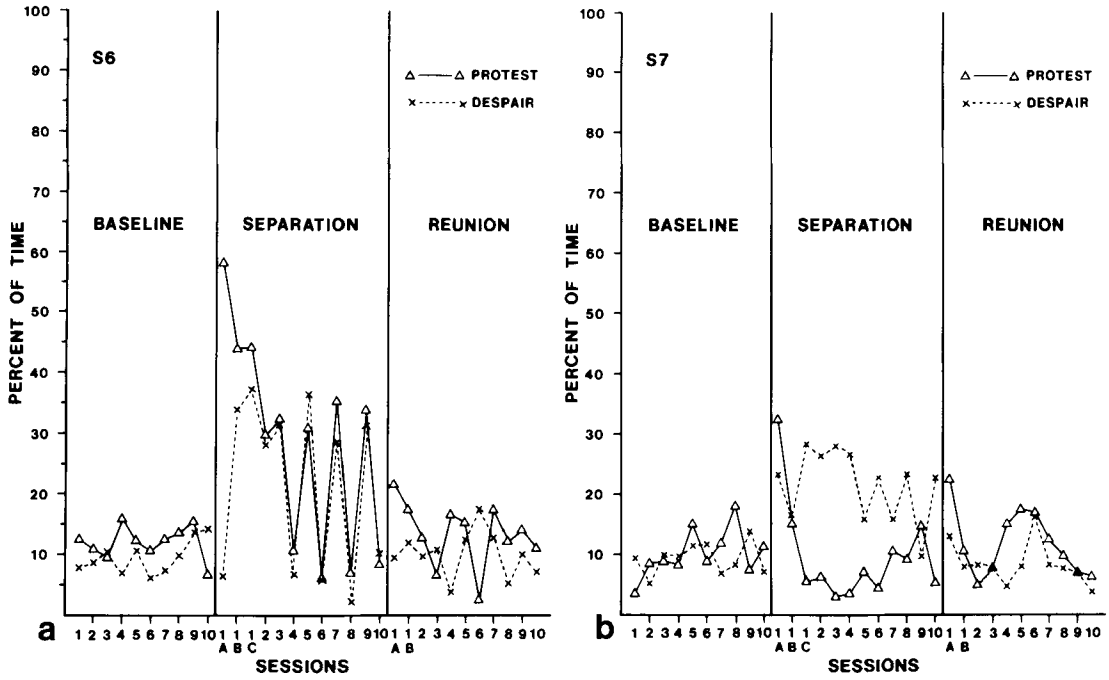


Fig. 3. The protest and despair reactions for Sara (a) and Nathan (b).

TABLE III. Significant Reactions Under Three Conditions of Separation*

	Days of separation											
Subjects	1A	1B	1C	2	3	4	5	6	7	8	9	10
Alone												
Tika	P	P	P	d	d	d	d	P	d	d	d	d
Arctica	[a]	P		A	P		P	dA	d	d	P	d
Joye	PA	P	—	d	d	dA	Pd	d	Pd	A		A
Artifact	PA	d	PA	PA	d	[a]A	d	A	Pd	dA	d	d
Sara	P	Pd	Pd	Pd	Pd		Pd		Pd		Pd	
Nathan	Pd	d	d	d	d	d	d	d	d	d		d
Together												
Sabrina	PC	P	—		d		A	[a]			d	A
Puddin	C	C	—		A						C	
Betsie	[a]		—		A	[a]		[a]	A	P		
Hallie	[a]C	d	—	A	A	A		A				
Reduced space												
Artica	AC	—	—		d	A				d		
Tika	C	—	—	A						[a]C	A	

*1A, 1B, 1C refer to the first three sessions of the first day, respectively. P = protest, d = despair, A = aggressive display, C = cling, [a] = protest without affective distress. When observations were not recorded for a session then a dash (—) is marked. See text for detailed explanations.

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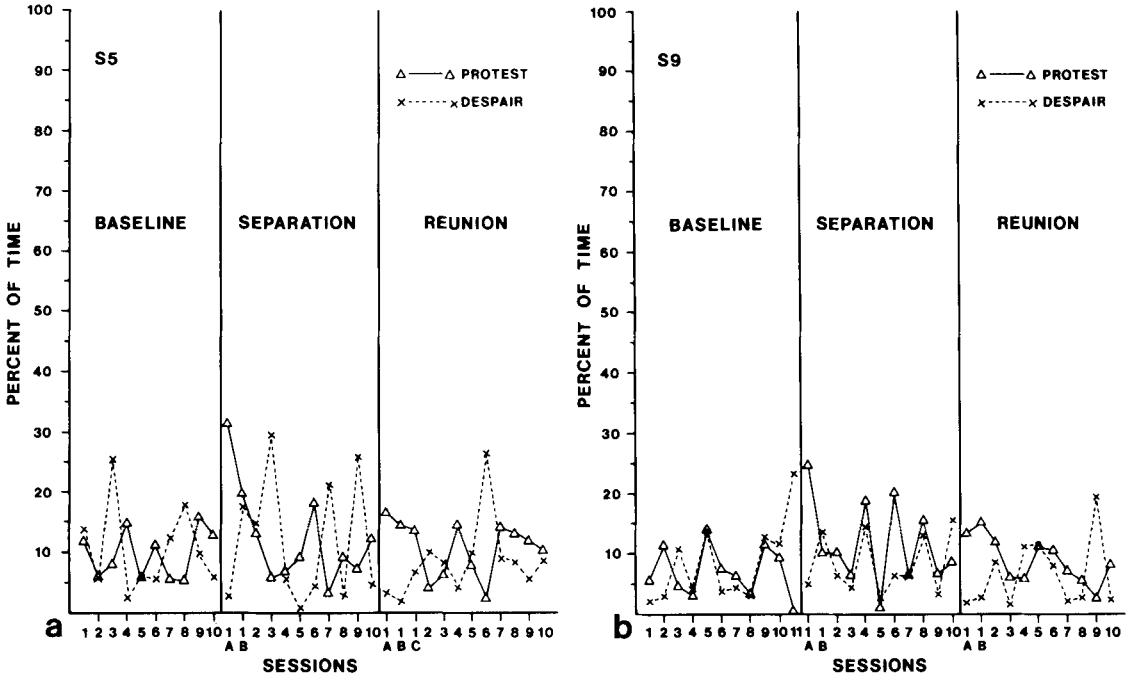


Fig. 4. Reaction to separation and reunion for Sabrina (a) and Betsie (b).

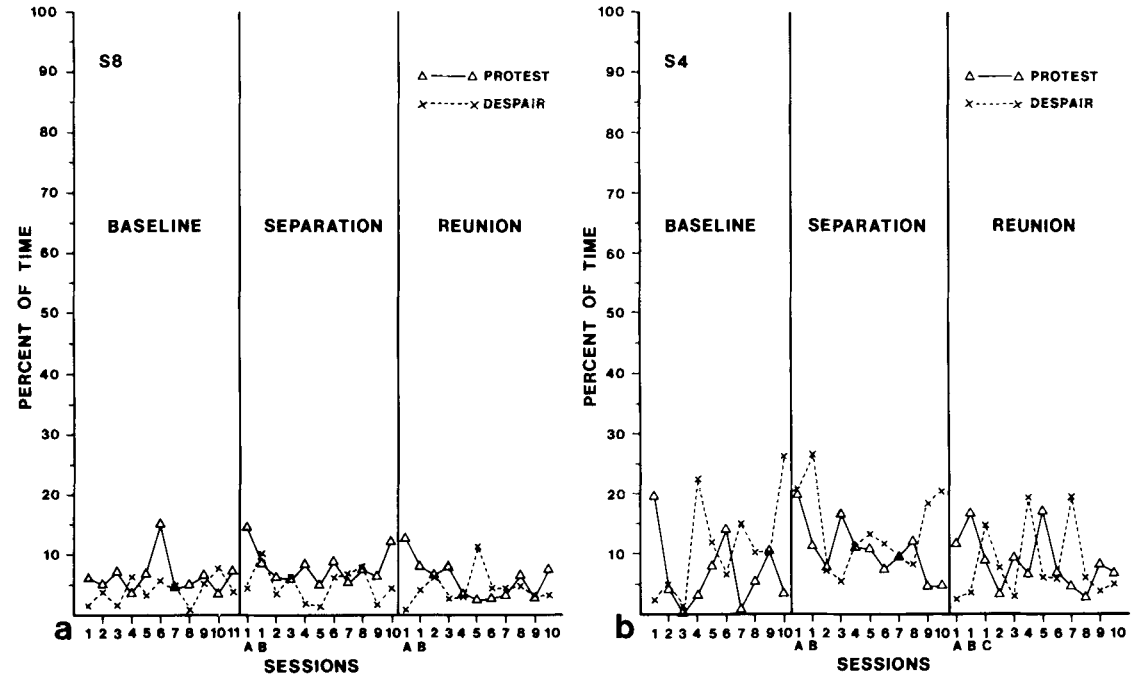


Fig. 5. Reaction to separation and reunion for Hallie (a) and Puddin (b).

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Additionally, five of six infants exhibited at least one recurrence of increased protest. While remaining significantly greater than baseline, the amount of protest for all subjects was less, subsequently, than that seen on the first day.

Days 2–10—together. Only two of the four chimpanzees who remained with a cagemate during separation showed any signs of either protest or despair after the first day. Protest alone was observed in one subject (Betsie), and only one subject (Sabrina) expressed both protest and despair (Table IV).

An analysis of social interaction during baseline between the partners of the original triads was undertaken in order to determine why only one member of each together group reacted to separation. The proportion of the social behaviors of play, look, and cling that were directed toward each member of the triad by the two other members demonstrated that the separated infant was the preferred partner for one of the two together infants (Fig. 6a,b). Protest and despair, therefore, occurred in the individual of each group that suffered the greater reduction in social contact, ie, whose preferred partner was on the opposite side of the barrier during separation.

Reaction During Reunion

First day. Immediately upon reunion, all the subjects clung to their cagemates. This effect was short-lived, however, since only two of the ten subjects were still clinging two minutes after reunion. When the first session was considered as a whole, only two of the ten chimpanzees demonstrated significantly increased amounts of clinging. Eight of ten subjects, however, showed significantly increased amounts of at least one social behavior during the first day of reunion (Table IV).

Six of the ten chimpanzees exhibited significant increases in certain dimensions of the protest category on the first day of reunion, but affective distress was associated only with the removal of the barrier. In the absence of distress, therefore, these measurements primarily reflected heightened arousal characterized by increased stereotypy or locomotion.

Prolonged reaction throughout reunion. There were no categories in which differences between baseline and reunion means were found for all subjects, nor for all of those who were alone, nor all of those who were together during separation. Those categories in which significant differences were found in at least half, of either the alone group or the together group, are presented in Table IV.

TABLE IV. Social Behaviors That Significantly Increased During Reunion*

Subject	Play	Look	Groom	T.W.	Cling
Alone					
Arctica		1A, 1B, R	1A, 1C, R	1B	
Tika		1B, R	1C	1A	1A R
Joye			R	R	
Artifact		1A	R		
Sara		1A, 1B, R		1A	
Nathan	R	1B		1A	
Together					
Sabrina	1A, 1B	1B			
Puddin					
Betsie		1A, 1B, R			1A, 1B, R
Hallie		1B, R			

*1A, 1B, and 1C refer to the first three sessions, respectively, of the first day of reunion. R refers to the entire reunion phase. T. W. = tandem walk. See text for further explanation.

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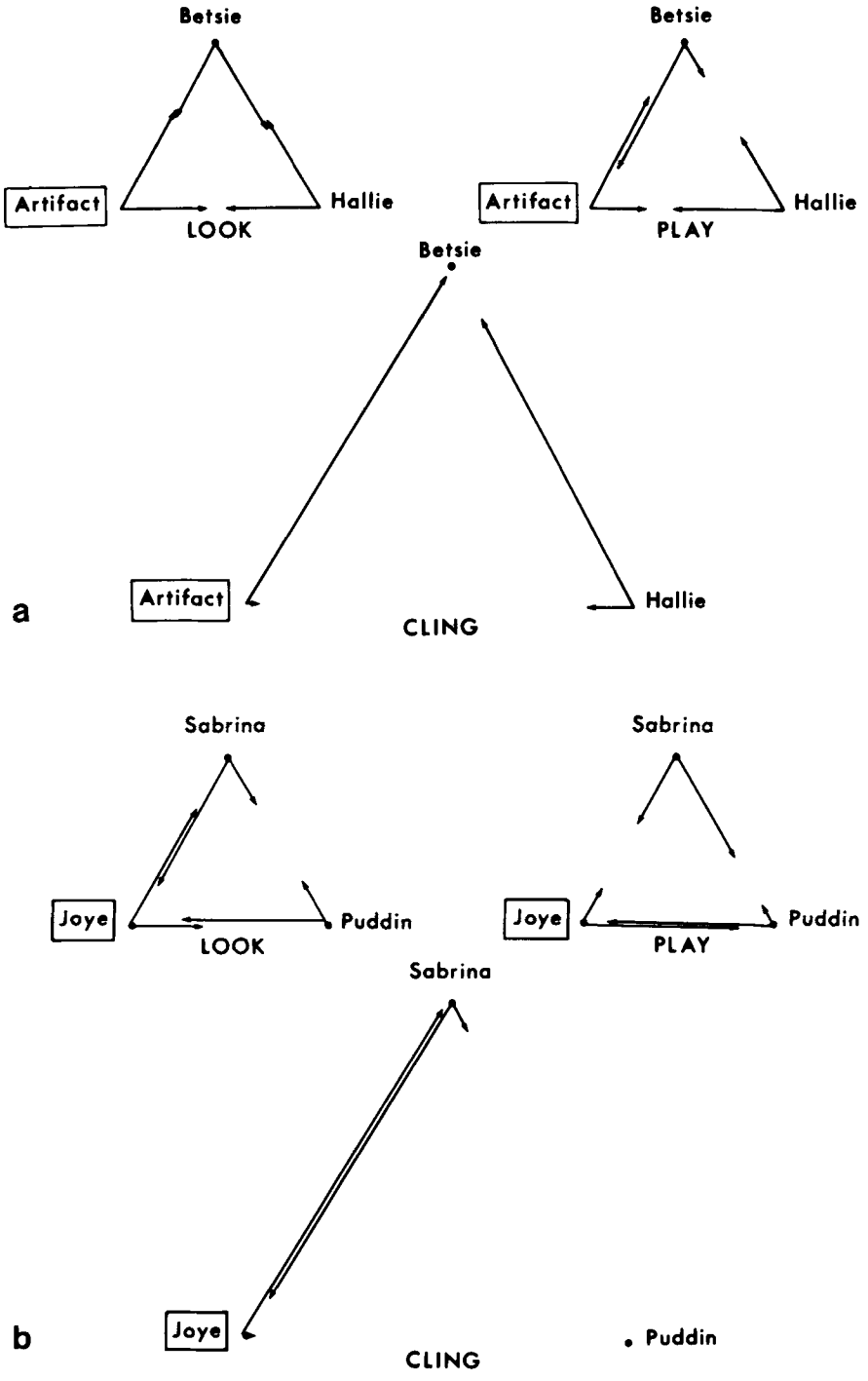


Fig. 6. Proportion of social behaviors directed toward each cagemate in the triad of Betsie, Hallie, and Artifact (a) and Sabrina, Puddin, and Joye (b). Length of line is approximate proportion in the direction of the arrow. Individual's name in box was alone during separation.

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These measures indicate that more time was spent engaged in positive social behaviors during reunion as compared with baseline. Half of the subjects spent more time looking at their cagemates, three subjects engaged in longer or more frequent bouts of grooming, and two individuals exhibited increased levels of clinging.

Reduced Space

The question of whether or not the effect of reducing the cage size with a metal partition was responsible for the occurrence of protest and/or despair during separation was specifically addressed by this additional phase. The results are presented in comparison with the separation groups in Table III. Although a significant amount of protest was seen in one subject, and significant amounts of despair were seen in the other subject, neither exhibited both protest and despair during the reduced space phase. Additionally, the patterning of the reaction was clearly different in this reduced space phase as compared with the pattern of high protest on the first day and despair on subsequent days, which was seen during separation.

DISCUSSION

The results of this study indicate that chimpanzees react in predictable ways to separation from cagemates. Protest is the initial response to separation from a primary attachment figure. Moreover, the chimpanzees of this study reacted in a manner in some ways similar to monkeys and in other respects more like children. Most chimpanzees began to protest immediately upon separation, which is characteristic of monkeys [Mineka & Suomi, 1978] and most children [Heinicke & Westheimer, 1965]. The chimpanzees had recurrences of protest throughout separation. This is similar to the protest reactions seen in children, which consists of repeated occurrences of crying and screaming during the first nine days of separation [Robertson, 1953; Heinicke & Westheimer, 1965]. In monkeys, on the other hand, protest reactions typically occur on the first two days and nights [Reite et al, 1981] and do not recur subsequent to despair [Mineka & Suomi, 1978].

The definitions used in this study differed somewhat from those used in studies of other species. Protest, in children, is defined by crying and attempts to regain contact with the mother [Bowlby, 1960]. In monkeys, protest is operationally defined as excessive vocalizations and disoriented, agitated locomotion [Minkea & Suomi, 1978]. In the present study, protest was defined by locomotion, stereotypy, and affective distress. Inclusion of stereotypy, as a category indicative of distress, was based on earlier studies of young nursery-reared chimpanzees [Davenport & Rogers, 1970]. Distress vocalizations and facial expressions do not occur randomly or in the absence of a stressful situation, and, as such, the presence of these characteristics is the most reliable and valid indicator of distress. Therefore, in the present study, a distinction was made between incidences of protest that included affective distress and those that did not.

The complete pattern of aggression in adult chimpanzees consists of a preliminary display, followed by physical aggression [Van Lawick-Goodall, 1971]. Immature components were used in this study as indicators of aggression and hostility. In some children, aggression was displayed to the visiting father, the care-giving nurses, and to siblings during separation from the mother [Heinicke & Westheimer, 1965; Robertson & Robertson, 1971]. In reunion, hostility was sometimes directed to the mother. The present conclusion is that the presence of increased levels of display during separation is similar to the frustration and hostility seen in children upon the loss of the attachment figure. Reunion with the partner resulted in decreased display, ie, decreased hostility and agression.

The reaction of chimpanzees to reunion appeared to be intermediate between monkeys and humans. While monkey infants show immediate clinging and long-term reintensification of ventroventral contact, human children demonstrate initial ambivalence

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to the mother and later demonstrate heightened vigilance and maintenance of proximity. For all the current subjects, the initial response was clinging, but in general, it was very brief. The removal of the barrier was unavoidably noisy and may have precipitated the cling response [eg see Willott & McDaniel, 1974]. There was, however, a prolonged increase in social contact, including increased looking at and grooming of their cagemate(s). There is a parallel between the chimpanzees' spending more time looking at their cagemates and the behavior of children who follow their mothers around the house, unwilling to lose sight of them. These findings suggest that the increased social behaviors evident during reunion reflected the chimpanzees' efforts to reestablish social relations with their attachment figures.

An investigation of each partner's participation in maintaining the relationship and an evaluation of the "security" of the attachment bond can facilitate an understanding of the reaction to social separation in chimpanzees. Within the triads, partner preference, ie, relative attachment, was determined by the amount of social interaction directed differentially toward each member. This showed that separation from a preferred partner resulted in a stronger reaction than separation from a nonpreferred one. An assessment of the quality of attachment, as was done for children utilizing the Ainsworth strange situation [Ainsworth & Bell, 1970], would enable classification of the type of attachment as either anxious or secure or avoidant. It is our belief that with a larger number of subjects, such a measurement could better elucidate the basis of variation between the subjects.

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